

Django Tutorial #22 - Create Model, Migrations & SQLite DB

Complete Handwritten Study Notes (Hinglish Language)

1. Django Model Kya Hota Hai?

Django Model ek Python class hoti hai jo database (DB) table ko represent karti hai. Yeh `models.py` file ke andar likhi jaati hai.

- **Table Representation:** Har ek Model class database ke andar ek individual table ko represent karti hai.
- **Columns & Fields:** Table ke andar jo columns bante hain (jaise name, age, email), unhe model ke andar **Fields** ke roop mein define kiya jata hai.
- **ORM Magic:** Aapko SQL queries likhne ki zaroorat nahi padti; Django aapke Python code ko automatic SQL query mein convert kar deta hai.

2. Model Create Karne Ka Tarika (`models.py`)

Apni app (jaise `blog` app) ke andar `models.py` file open karein aur is tarah code likhein:

```
from django.db import models class Student(models.Model): name = models.CharField(max_length=100) age = models.IntegerField() email = models.EmailField(unique=True) enrollment_date = models.DateField(auto_now_add=True)
```

3. Important Fields Description

CharField

Text ya naam store karne ke liye. Isme `max_length=100` dena compulsory hota hai.

IntegerField

Numbers ya age jaisi numeric values ko store karne ke liye use hota hai.

EmailField

Email validate karne ke liye. `unique=True` se duplicate emails allow nahi hoti.

DateField

Date store karne ke liye. `auto_now_add=True` se current date auto-save ho jati hai.

4. Migrations aur Database Commands

Model likhne ke baad table direct database mein nahi banti. Uske liye do main terminal commands run karni padti hain:

Step 1: Make Migrations Command

```
python3 manage.py makemigrations
```

Yeh command aapke Python code ko read karke **migrations** folder ke andar ek initial file create karti hai (jaise **0001_initial.py**), jisme SQL queries hoti hain.

Step 2: Migrate Command

```
python3 manage.py migrate
```

Yeh actual command database (**db.sqlite3**) ke andar tables ko create aur update karti hai. Pehli baar chalane par Django default internal tables ke sath aapki nayi table bhi bana deta hai.

5. Database Check Karna (SQLite DB)

- SQLite database ke andar table create hui ya nahi, yeh check karne ke liye VS Code mein **SQLite Viewer / SQLite Editor** extensions ka use kar sakte hain.
- Table ke andar columns (**id** , **name** , **age** , **email** , **enrollment_date**) proper primary key aur constraints ke sath properly ban jaate hain.
- Jab bhi aap models mein koi change karte hain, dobara **makemigrations** aur **migrate** chalane par Django ek nayi migration file banakar changes ko database mein reflect kar deta hai.

Next Lecture Preview

Aane wale lectures mein hum seekhenge ki is Model/Table ke andar data kaise insert, update, modify aur delete karte hain (**CRUD Operations**).